Mutex Stress Test (RAM-Resident Execution) on Raspberry Pi Pico 2 (RP2350)

This directory contains the RAM-resident variant of the mutex stress test suite for the Raspberry Pi Pico 2 (RP2350, dual Cortex-M33).

Like `cmsis-os-validator-ram`, this test is linked to execute **entirely from internal SRAM** (`0x20000000`) without writing anything to the external QSPI flash.

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1. Overview & Architecture

- **Target Architecture**: Raspberry Pi Pico 2 (RP2350, 2x Cortex-M33).
- **Toolchain**: `arm-none-eabi-gcc` 15.2 (xPack GNU Arm Embedded Toolchain).
- **Debugger**: OpenOCD via CMSIS-DAP (Raspberry Pi Debug Probe).
- **Execution Target**: Internal SRAM (`0x20000000`, 520 KB).

Isolation Guarantee

- All modifications are strictly isolated to this test directory and its test hook registrations.
- Existing flash-based tests, board-level OpenOCD configs, and framework boot flows remain completely unmodified.

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2. Components

File	Purpose
`linker-ram.ld`	Dedicated linker script mapping all loadable sections (`.text`, `.rodata`, `.data`) to SRAM `0x20000000`. Marks `.bss`, `.heap`, `.stack` as `(NOLOAD)` and embeds Pico

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	`.embedded_end_block` to keep the download size compact.
`harness-suite.cpp`	Test registration stub including `../harness-suite.hpp` for `__wrap_os_main()` semihosting and UART verdict reporting.
`hw.sh`	Custom hardware test runner. Sets OpenOCD target work-area to high SRAM (`0x20070000`), automates direct SRAM image loading, core 1 bootrom holding, core 0 VTOR/SP/PC configuration, semihosting monitoring, and clean exit.
`load-ram.cfg`	Standalone TCL script for interactive debugging or manual OpenOCD flashing.

3. Running the Test

Via CTest (Micro-OS+ SMP Test Framework)

```
cd /home/dan/Work/micro-os-plus-iii-smp/tests/build/cortexm-pico2-
        cmake-gcc-release
ctest -V -R cortexm-pico2-mutex-stress-ram-hwd
```

Via Standalone Script

```
/home/dan/Work/micro-os-plus-iii-cortexm/test/boards/pico2/hw.sh
        mutex-stress-ram
```